

PARUL UNIVERSITY
FACULTY OF ENGINEERING & TECHNOLOGY
B.Tech. Summer 2023- 24 Examination

Semester: 05**Subject Code: 203105318****Subject Name: Design and Analysis of Algorithm****Date: 25/04/2024****Time: 10:30am to 1:00pm****Total Marks: 60****Instructions:**

1. All questions are compulsory.
2. Figures to the right indicate full marks.
3. Make suitable assumptions wherever necessary.
4. Start new question on new page.

Q.1	Objective Type Questions - (Fill in the blanks, one word answer, MCQ-not more than Five in case of MCQ) (All are compulsory) (Each of one mark)	(15)	CO	PO	Bloom's Taxonomy
	1.What is the time complexity of binary search in a sorted array of size n? A) $O(n)$ B) $O(\log n)$ C) $O(n^2)$ D) $O(1)$		1	1	Analyzing
	2.Which sorting algorithm is known for its stability? A) Quick Sort B) Merge Sort C) Heap Sort D) Bubble Sort		3	3	Understanding
	3.Which of the following data structures follows the Last In First Out (LIFO) principle? A) Queue B) Stack C) Linked list D) Heap		3	3	Understanding
	4.Which algorithm is used to solve the knapsack problem? A) Greedy algorithm B) Dynamic programming C) Backtracking D) Divide and conquer		3	3	Understanding
	5.Which algorithm is used to solve the traveling salesman problem? A) Greedy algorithm B) Dynamic programming C) Backtracking D) Branch and bound		3	3	Understanding
	6.Define dynamic programming.		4	5	Evaluating
	7.What is the principle of optimality?		1	1	Analyzing
	8.Define recurrence relation.		4	5	Evaluating
	9.Explain the concept of greedy choice property.		1	2	Analyzing
	10.Define Big O notation.		4	5	Evaluating
	11.What is the main advantage of using Divide and Conquer technique?		1	1	Analyzing
	12.Define Backtracking.		4	5	Evaluating
	13.What is the time complexity of linear search in an array of size n?		1	1	Analyzing
	14.Define pseudo-code.		4	5	Evaluating
	15.Matrix $A(2 \times 3)$, $B(3 \times 2)$ and $C(2 \times 2)$. Find the dimensions of ABC.		5	4	Evaluating

Q.2	Answer the following questions. (Attempt any three)	(15)			
	A) Explain the working principle of Merge Sort algorithm and analyze its time complexity.		1	2	Analyzing
	B) Describe Prim's algorithm for finding the minimum spanning tree of a graph.		4	5	Evaluating
	C) For the recurrence relation $T(n) = 2T(n/2) + n$, solve for $T(n)$ and analyze its time complexity.		1	2	Analyzing
	D) If an algorithm has a time complexity of $O(n^2)$ and requires 100 operations for an input size of 10, what would be the expected number of operations for an input size of 100?		4	5	Evaluating
Q.3	A) Simulate Merge sort algorithm for the following example 25,36,12,4,5,16,58,54,24,16,9,65,78	(07)	5	4	Understanding
	B) Assuming that a hash table has 11 items, use a simple function $\text{hash}(\text{key}) = \text{key} \bmod 11$ and map the following keys : a) 12, 56, 78, 67, 99, 28, 97 b) Is there any collision?	(08)	5	4	Understanding
	OR				
	B) How many solutions are possible for a 4-Queen problem?	(08)	4	4	Analyzing
Q.4	A) Find the sum of subsets for the following set of integers. [1 2 3 5 6 7 8 9 10] for $w = 8$.	(07)	5	4	Analyzing
	OR				
	A) Construct a BST for the following set of numbers. 10, 6, 7, 12, 14, 43	(07)	2	1	Analyzing
	B) Find LCS between strings A and B. A= EXAMPLE B=APE	(08)	2	1	Understanding